Part 3: Market Failures Practice Problems

Econ 502: Advanced Microeconomics

1. Adverse Selection in Insurance with Two Types

A population of consumers each has wealth $w = 100$. With probability p_i, they suffer a loss of $L = 36$. Consumers have utility $u(z) = \sqrt{z}$ and so are risk-averse. There are two risk types:

- **Low-risk** consumers have $p_L = 0.2$
- **High-risk** consumers have $p_H = 0.6$

A fraction λ of the population is high-risk. Each consumer knows their own type; the insurer cannot tell types apart. Insurance is "full insurance" at premium P (the insurer pays L if the loss occurs).

a) For each type $i \in \{L, H\}$, find the **certainty equivalent** CE_i of facing the loss without insurance. Use the certainty equivalent to compute the maximum premium $P_i^{\max}$ that makes the consumer just indifferent between buying full insurance and going uninsured. Verify that $P_H^{\max} > P_L^{\max}$.

b) Suppose the insurer charges the **actuarially fair pooled premium**:

$$P_{\text{pool}} = [\lambda p_H + (1 - \lambda)p_L] \cdot L$$

For $\lambda = 0.5$, compute P_{pool}. Which types are willing to buy at this premium? Will the insurer break even?

c) For what range of λ does the market unravel (low-risk types drop out, leaving only high-risk types)? Explain the economic logic in words.

d) Suppose the government mandates that everyone purchase insurance at the pooled premium P_{pool} with $\lambda = 0.5$. Compare each type's expected utility under the mandate with their expected utility without insurance. Who is better off? Who is worse off? Why might this still be a desirable policy from a social welfare perspective?

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2. Pigouvian Tax and Cap-and-Trade with Heterogeneous Firms

Two firms produce a chemical, with pollution as a byproduct. Firm i's private marginal cost of producing q_i units is:

$$MC_1(q_1) = 2q_1, \qquad MC_2(q_2) = 4q_2$$

The market price is $P = 40$ per unit. Each unit of output produces 1 unit of pollution, and each unit of pollution causes \$8 of damage to a downstream community.

a) Find each firm's privately optimal output q_i^{priv} (where $P = MC_i$). Compute total output and total pollution damage.

b) Find the socially efficient output q_i^* for each firm (where $P = MC_i + 8$). Compute total output, total damage, and the deadweight loss from unregulated production.

c) **Pigouvian tax.** The regulator imposes a per-unit tax t on output of both firms. Find the tax rate that decentralizes the efficient outcome and verify the equilibrium quantities.

d) **Cap-and-trade.** The regulator instead issues permits equal to the efficient total emissions from part (b) and distributes them equally between the two firms. Plants can trade permits at price τ. Each firm chooses q_i to maximize $Pq_i - C_i(q_i) - \tau(q_i - \bar{e}_i)$, where $\bar{e}_i$ is its initial allocation. Find the equilibrium permit price τ^* and the equilibrium quantities. Which firm buys, which sells, and how many permits change hands? How does the equilibrium compare to part (c)?

3. Coase Theorem with Reciprocal Externalities

A bakery and a doctor share a building. The bakery's machinery generates noise. Profits depend on whether the bakery uses noisy or quiet machinery and whether the doctor's office is soundproofed:

	Doctor not soundproofed	Doctor soundproofed
Bakery noisy	$\pi_B = 200, \pi_D = 60$	$\pi_B = 200, \pi_D = 100$
Bakery quiet	$\pi_B = 150, \pi_D = 130$	$\pi_B = 150, \pi_D = 100$

Soundproofing costs the doctor \$30 (already netted out of π_D in the soundproofed column). Switching to quiet machinery costs the bakery \$50 in foregone profit.

a) Compute total surplus $\pi_B + \pi_D$ for each cell. Which combination is socially efficient?

b) **Right to silence.** Suppose the doctor has the legal right to a quiet office. Without bargaining, what does the bakery do, and does the doctor soundproof? With Coasian bargaining, what is the efficient outcome and what is the range of feasible payments from bakery to doctor?

c) **Right to be noisy.** Suppose the bakery has the legal right to make noise. Without bargaining, what does the doctor do? Is bargaining required to reach the efficient outcome?

d) Compare your answers in (b) and (c). What does this illustrate about the Coase theorem? Why might the assignment of property rights still matter in practice?

Public Goods

4. Public Good Provision with Three Contributors

Three neighbors share a small park. Each has income $Y_i = 60$ and chooses how much to contribute $g_i \geq 0$ to park maintenance. Total park quality is $G = g_1 + g_2 + g_3$. Each neighbor's utility is:

$$U_i(x_i, G) = x_i \cdot G$$

where $x_i = Y_i - g_i$ is private consumption.

a) Derive each neighbor's best response function $g_i^*(g_{-i})$, where $g_{-i} = \sum_{j \neq i} g_j$. By symmetry, find the Nash equilibrium contributions g_i^* and total provision G^*.

b) **Social optimum.** A planner chooses (x_1, x_2, x_3, G) to maximize $\sum_i U_i$ subject to $\sum_i x_i + G = \sum_i Y_i$. Find G^{**} and the equal-private-consumption allocation x_i^{**}. (Hint: with utility $x_i G$, total welfare is XG where $X = \sum x_i$.)

c) Verify that the Nash equilibrium satisfies $MRS_i = 1$ for each i but the social optimum satisfies $\sum_i MRS_i = 1$ (the **Samuelson condition**). Compute the ratio G^*/G^{**}. As the number of contributors N grows, what happens to the fraction of the efficient public good that is voluntarily provided?

d) Suppose each contributor receives a subsidy s on their contribution: a contribution of g_i costs them only $(1-s)g_i$, with the government covering the rest. Find the subsidy rate s^* that decentralizes the efficient public good level G^{**}.

5. Lindahl Prices

Two consumers fund a public good G. They have utility:

$$U_1(x_1, G) = x_1 + 10 \ln G, \qquad U_2(x_2, G) = x_2 + 6 \ln G$$

Each unit of G costs \$1 to produce (so the resource constraint is $x_1 + x_2 + G = Y_1 + Y_2$). Both have ample income.

a) **Samuelson condition.** Compute each consumer's $MRS_i = (\partial U_i/\partial G)/(\partial U_i/\partial x_i)$ and find the efficient G^{**}.

b) **Lindahl prices.** Suppose each consumer is charged a personalized price-per-unit τ_i for the public good, where $\tau_1 + \tau_2 = 1$ (the prices add up to the marginal cost of G). Each consumer takes τ_i as given and chooses how much G they would like to "demand," then the level of G is set so both demand the same quantity. Find the Lindahl prices (τ_1^*, τ_2^*) that support the efficient G^{**}.

c) Verify that at the Lindahl prices, each consumer "demands" exactly G^{**}. Why does this mechanism elicit efficient provision in theory?

d) Why is implementing Lindahl pricing difficult in practice? What incentive do consumers have when reporting their preferences to the planner?